Third Workshop on Augmented Intelligence in Technology-Assisted Review Systems (ALTARS)

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Abstract. In this third edition of the workshop on Augmented Intelligence in Technology-Assisted Review Systems (ALTARS), we focus on the capacity of building test collections for the evaluation of High-recall Information Retrieval (IR) systems which tackle challenging tasks that require the finding of (nearly) all the relevant documents in a collection. During the workshop, the organizers as well as the participants will discuss the problems of how to build and evaluate these types of systems and prepare a set of guidelines for a correct evaluation of those systems according to the current and future available datasets.

Keywords: Technology-Assisted Review Systems · Augmented Intelligence · Evaluation · Systematic Reviews · eDiscovery

1 Introduction

Technology-assisted review (TAR) systems use a kind of human-in-the-loop approach where classification and/or ranking algorithms are continuously trained according to the relevance feedback from expert reviewers, until a substantial number of the relevant documents are identified. This approach has been shown to be more effective and more efficient than traditional e-discovery and systematic review practices, which typically consists of a mix of keyword searches and manual review of the search results.

The first edition of the ALTARS workshop was successfully held at ECIR 2022[1] [5]. During (and after) that workshop, there was a lively discussion about open questions that still need to be addressed and clarified in this research area. The second edition witnessed an even larger participation at ECIR 2023[2] [6]

[1] https://altars2022.dei.unipd.it/.
[2] https://altars2023.dei.unipd.it/.

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with around 20 participants on-site and 10 online. The aim of the second edition was to study innovative approaches to fathom the effectiveness of these TAR systems.

In this third edition of the workshop[3], we will invite researchers to discuss the issues related to the shortage of test collections for the evaluation of TAR systems and how to facilitate standardized evaluations and the creation of such datasets and to ensure that TAR systems are fine-tuned to meet the high-recall retrieval requirements of tasks like electronic discovery, systematic reviews, and investigations.

2 Workshop Goals and Objectives

The main goal of this edition of the workshop will be to focus on the evaluation of the different definitions of the effectiveness of TAR systems which is a research challenge itself. The idea is to go beyond a "traditional" retrieval approach and study the problem from different perspectives.

In the first edition of the workshop, we organized a special issue for the Intelligent Systems with Applications journal where extended versions of the papers presented at the workshop were published in open access [4].[4]

In the second edition of the workshop, we organized another special issue for the MDPI Information journal with the same aim (call for papers is still ongoing at present time).[5]

The desired outcome of this third workshop is having a collection of high-quality short papers to be presented at the workshop and a selected number of papers that will be invited in a new special issue and gather momentum in this interdisciplinary area together with researchers, stakeholders in the fields (such as the https://icasr.github.io/about.html and https://www.zylab.com/en/), and international projects (https://dossier-project.eu).

The goals of the workshop are threefold:

- To foster cross-discipline collaborations between researchers with different perspectives and research backgrounds in the TAR systems;
- To combine and analyze existing theoretical and empirical contributions in order to determine shared issues, and novel research questions;
- To create a set of shared datasets dedicated towards the evaluation of TAR systems, thus enabling a wider research community to benefit from the outcomes of the workshop.

2.1 Challenges and Issues

One last goal of the workshop is to discuss open issues and challenges and have as a final product a Horizon Europe/NSF proposal to strengthen the network of researchers in this topic.

[3] https://altars2024.dei.unipd.it/.
[4] https://www.sciencedirect.com/journal/intelligent-systems-with-applications/special-issue/10CLX27456Q.
[5] https://www.mdpi.com/journal/information/special_issues/74F9Z52LA4.

In particular, in the first two editions of the ALTARS workshop, the discussions have focused on the challenges that reflect the complexity and diverse applications of TAR systems in contexts where high recall is crucial. These issues show the multifaceted nature of research and development in this field:

– Scalability: Developing TAR systems that can efficiently handle large volumes of diverse data, especially in e-Discovery where various document types and formats need classification.
– Effectiveness: Achieving high recall without compromising precision remains a challenge, particularly in contexts like systematic reviews where the comprehensive gathering of relevant studies is crucial.
– Algorithmic Bias: Addressing biases in test collections due to missing relevant labels is essential. Developing methods to mitigate biases and ensure representative test collections for fair evaluation of retrieval models.
– Varied Data Types: Creating TAR systems that can effectively process and interpret different types of data, including text, images, and structured data, as seen in e-Discovery where correspondence, memos, and balance sheets need classification.
– Human-in-the-Loop: Designing systems that incorporate human expertise effectively, especially in contexts like systematic reviews where domain knowledge is crucial in identifying relevant studies.
– Generalizability: Ensuring the applicability and effectiveness of TAR systems across diverse domains, considering that systematic reviews, e-Discovery, and other contexts may have distinct requirements and nuances.
– Evaluation Metrics: Developing robust evaluation metrics and standards to assess the performance of TAR systems accurately, especially when dealing with complex tasks like systematically gathering evidence or classifying legal documents.
– Ethical and Legal Implications: Addressing ethical concerns and legal implications, particularly in e-Discovery, where the accuracy and fairness of document classification can impact legal proceedings.

3 Format and Structure

This workshop will be a full-day workshop and will be structured in four sessions: two sessions in the morning and two in the afternoon.

The call for papers of the workshop will include both full and short papers. All the authors of the accepted papers will have the possibility to give a talk; moreover, during an afternoon coffee break, we plan to organize a poster session to have some additional time for extra discussions that may arise during the day.

There will be an invited keynote speaker in the morning (to be confirmed) and a panel/general discussion in the afternoon after all the paper presentation.

4 Organizing Team

All the three organizing committee members have been active participants in the past editions of the TREC, CLEF and FIRE evaluation forum for the Total

Recall and Precision Medicine TREC Tasks, TAR in eHealth tasks, and AI for Legal Assistance. The committee members have strong research record with a total of more than 400 papers in international journals and conferences. They have been doing research in technology-assisted review systems and problems related to document distillation both in the eHealth and eDiscovery domain and made significant contributions in this specific research area.

Giorgio Maria Di Nunzio is Associate Professor at the Department of Information Engineering of the Universiryt of Padova. He has been the co-organizer of the ongoing Covid-19 Multilingual Information Access Evaluation forum,[6] in particular for the evaluation of high-recall systems and high-precision system tasks. He will bring to this workshop the perspective of alternative (to the standard) evaluation measures and multilingual terminological challenges [2,3,7,8,11].

Evangelos Kanoulas is Full Professor at the Faculty of Science of the Informatics Institute at the University of Amsterdam. He has been the co-organizer CLEF eHealth Lab and of the Technologically Assisted Reviews in Empirical Medicine task.[7] He has also been the co-organizer of a related workshop at ECIR 2023, the first international workshop on Legal Information Retrieval (LegalIR). He will bring to the workshop the perspective of the evaluation of the costs in eHealth TAR systems, in particular of the early stopping strategies as well as the conversational aspects of interactive systems [10,13,14].

Prasenjit Majumder is Associate Professor at the Dhirubhai Ambani Institute of Information and Communication Technology (DA-IICT), Gandhinagar and TCG CREST, Kolkata, India. He has been the co-organizer of the Forum for Information Retrieval Evaluation and, in particular, the Artificial Intelligence for Legal Assistance (AILA) task.[8] He will bring to the workshop the perspective of the evaluation of the costs of eDiscovery, in particular of the issues related to legal precedence findings and text representations in financial texts [1,9,12].

References

1. Bhattacharya, P., et al.: FIRE 2020 AILA track: artificial intelligence for legal assistance. In: Majumder, P., Mitra, M., Gangopadhyay, S., Mehta, P. (eds.) Forum for Information Retrieval Evaluation, FIRE 2020, Hyderabad, India, 16–20 December 2020, pp. 1–3. ACM (2020). https://doi.org/10.1145/3441501.3441510
2. Clipa, T., Di Nunzio, G.M.: A study on ranking fusion approaches for the retrieval of medical publications. Information **11**(2), 103 (2020). https://doi.org/10.3390/info11020103
3. Di Nunzio, G., Faggioli, G.: A study of a gain based approach for query aspects in recall oriented tasks. Appl. Sci. **11**(19), 9075 (2021). https://doi.org/10.3390/app11199075. https://www.mdpi.com/2076-3417/11/19/9075

[6] http://eval.covid19-mlia.eu.
[7] https://clefehealth.imag.fr.
[8] https://sites.google.com/view/aila-2021.

4. Di Nunzio, G.M., Kanoulas, E.: Special issue on technology assisted review systems. Intell. Syst. Appl. **20**, 200260 (2023). https://doi.org/10.1016/j.iswa.2023.200260. https://www.sciencedirect.com/science/article/pii/S2667305323000856
5. Di Nunzio, G.M., Kanoulas, E., Majumder, P.: Augmented intelligence in technology-assisted review systems (ALTARS 2022): evaluation metrics and protocols for ediscovery and systematic review systems. In: Hagen, M., et al. (eds.) ECIR 2022. LNCS, vol. 13186, pp. 557–560. Springer, Cham (2022). https://doi.org/10.1007/978-3-030-99739-7_69
6. Di Nunzio, G.M., Kanoulas, E., Majumder, P.: 2nd workshop on augmented intelligence in technology-assisted review systems (ALTARS). In: Kamps, J., et al. (eds.) Advances in Information Retrieval - 45th European Conference on Information Retrieval, ECIR 2023, Dublin, Ireland, 2–6 April 2023, Proceedings, Part III. LNCS, vol. 13982, pp. 384–387. Springer, Switzerland (2023). https://doi.org/10.1007/978-3-031-28241-6_41
7. Di Nunzio, G.M., Marchesin, S., Silvello, G.: A systematic review of automatic term extraction: what happened in 2022? Digit. Scholarsh. Humanit. **38**(Suppl._1), 41–47 (2023). https://doi.org/10.1093/llc/fqad030
8. Di Nunzio, G.M., Vezzani, F.: Did I miss anything? A study on ranking fusion and manual query rewriting in consumer health search. In: Barrón-Cedeño, A., et al. (eds.) Experimental IR Meets Multilinguality, Multimodality, and Interaction - 13th International Conference of the CLEF Association, CLEF 2022, Bologna, Italy, 5–8 September 2022, Proceedings. LNCS, vol. 13390, pp. 217–229. Springer, Cham (2022). https://doi.org/10.1007/978-3-031-13643-6_17
9. Gangopadhyay, S., Majumder, P.: Text representation for direction prediction of share market. Exp. Syst. Appl. **211**, 118472 (2023). https://doi.org/10.1016/j.eswa.2022.118472
10. Li, D., Kanoulas, E.: When to stop reviewing in technology-assisted reviews: sampling from an adaptive distribution to estimate residual relevant documents. ACM Trans. Inf. Syst. **38**(4), 41:1–41:36 (2020). https://doi.org/10.1145/3411755
11. Marchesin, S., Di Nunzio, G.M., Agosti, M.: Simple but effective knowledge-based query reformulations for precision medicine retrieval. Inf. **12**(10), 402 (2021). https://doi.org/10.3390/info12100402
12. Parikh, V., et al.: AILA 2021: shared task on artificial intelligence for legal assistance. In: Ganguly, D., Gangopadhyay, S., Mitra, M., Majumder, P. (eds.) Forum for Information Retrieval Evaluation, FIRE 2021, Virtual Event, India, 13–17 December 2021, pp. 12–15. ACM (2021). https://doi.org/10.1145/3503162.3506571
13. Zou, J., Aliannejadi, M., Kanoulas, E., Pera, M.S., Liu, Y.: Users meet clarifying questions: toward a better understanding of user interactions for search clarification. ACM Trans. Inf. Syst. **41**(1), 16:1–16:25 (2023). https://doi.org/10.1145/3524110
14. Zou, J., Kanoulas, E.: Towards question-based high-recall information retrieval: locating the last few relevant documents for technology-assisted reviews. ACM Trans. Inf. Syst. **38**(3), 27:1–27:35 (2020). https://doi.org/10.1145/3388640